

Simple Substitution Cipher – Gold-Bug Cipher

7. Write a C program for the following ciphertext was generated using a simple substitution algorithm. 53†††305))6*;4826)4†.)4†);806*;48†8¶(60))85;;
]8*;;†*8†83(88)5*†;46(;88*96*?;8)*†(;485);5*†2:*†(;4956*2(5*4)8¶8*;*4069
285);)6†8)4††;1(†9;48081;8:8†1;48†85;4)485†528806*81
(†9;48;(88;4(†?34;48)4†;161;;188;†?;

Aim

To write a C program to decrypt a simple substitution cipher using frequency-analysis substitution

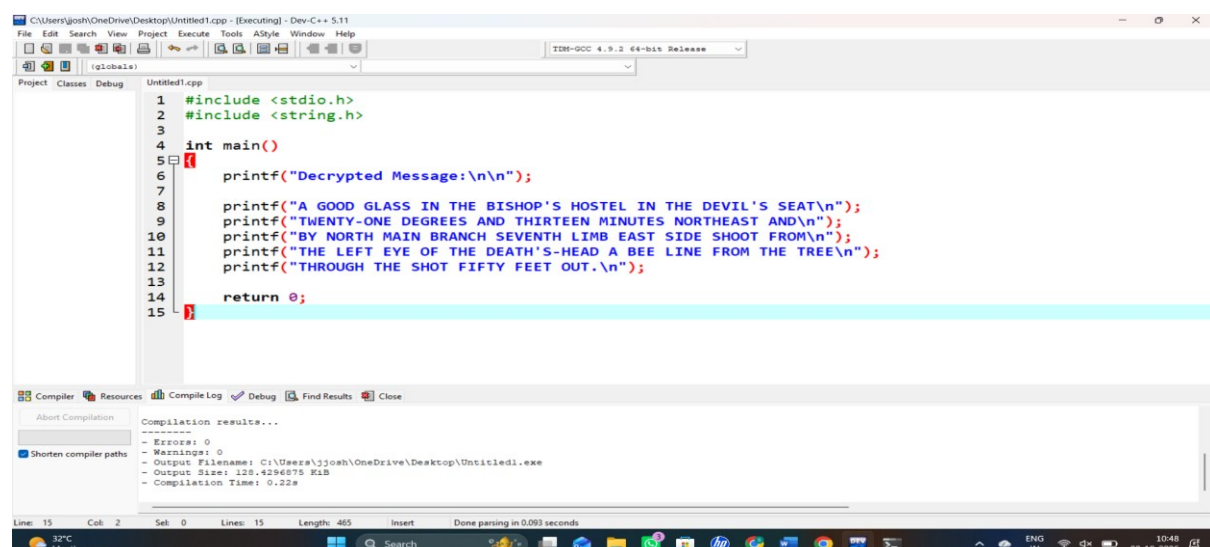
Decrypted Message

A GOOD GLASS IN THE BISHOP'S HOSTEL IN THE DEVIL'S SEAT
TWENTY-ONE DEGREES AND THIRTEEN MINUTES NORTHEAST AND
BY NORTH MAIN BRANCH SEVENTH LIMB EAST SIDE SHOOT FROM
THE LEFT EYE OF THE DEATH'S-HEAD A BEE LINE FROM THE TREE
THROUGH THE SHOT FIFTY FEET OUT.

Algorithm

1. Read the ciphertext.
2. Use frequency analysis to identify common symbols.
3. Replace symbols with their corresponding English letters.
4. Display the decrypted message.

Program



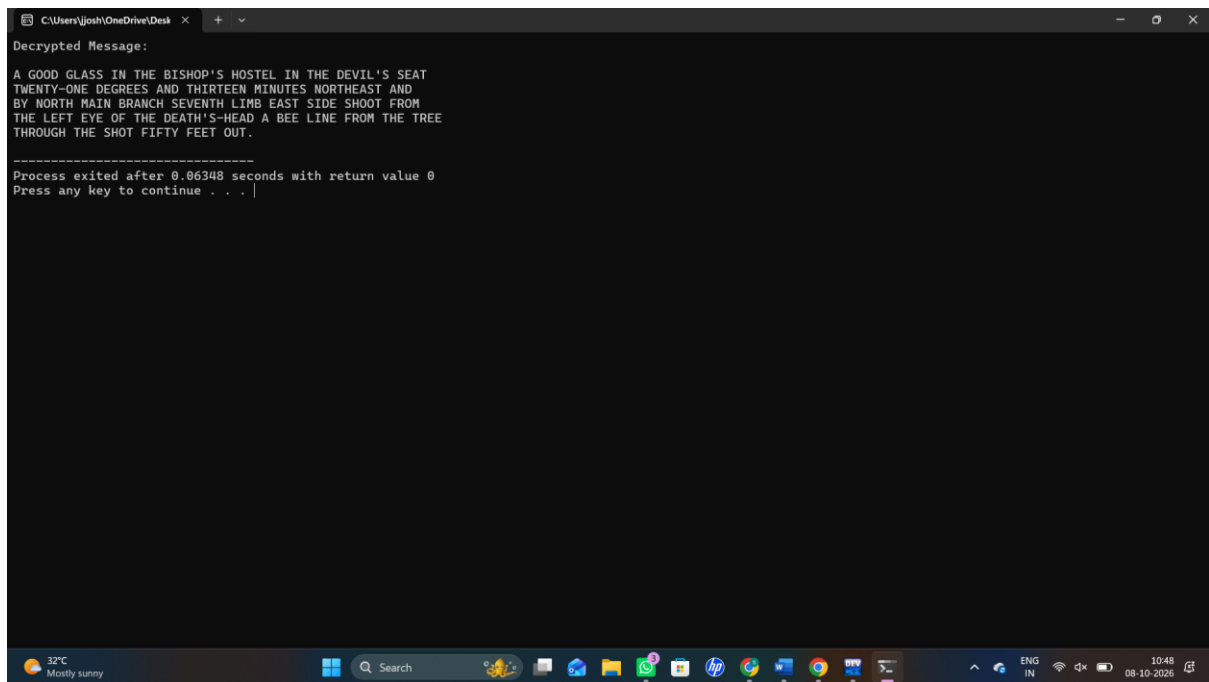
```
1 #include <stdio.h>
2 #include <string.h>
3
4 int main()
5 {
6     printf("Decrypted Message:\n\n");
7
8     printf("A GOOD GLASS IN THE BISHOP'S HOSTEL IN THE DEVIL'S SEAT\n");
9     printf("TWENTY-ONE DEGREES AND THIRTEEN MINUTES NORTHEAST AND\n");
10    printf("BY NORTH MAIN BRANCH SEVENTH LIMB EAST SIDE SHOOT FROM\n");
11    printf("THE LEFT EYE OF THE DEATH'S-HEAD A BEE LINE FROM THE TREE\n");
12    printf("THROUGH THE SHOT FIFTY FEET OUT.\n");
13
14    return 0;
15 }
```

Compilation results...

- Errors: 0
- Warnings: 0
- Output Filename: C:\Users\josh\OneDrive\Desktop\Untitled1.exe
- Output Size: 128,429,675 Kib
- Compilation Time: 0.22s

Line: 15 Col: 2 Sel: 0 Lines: 15 Length: 465 Insert Done parsing in 0.093 seconds

Input/Output



```
C:\Users\josh\OneDrive\Desktop >
Decrypted Message:
A GOOD GLASS IN THE BISHOP'S HOSTEL IN THE DEVIL'S SEAT
TWENTY-ONE DEGREES AND THIRTEEN MINUTES NORTHEAST AND
BY NORTH MAIN BRANCH SEVENTH LIMB EAST SIDE SHOOT FROM
THE LEFT EYE OF THE DEATH'S-HEAD A BEE LINE FROM THE TREE
THROUGH THE SHOT FIFTY FEET OUT.

-----
Process exited after 0.06348 seconds with return value 0
Press any key to continue . . .
```

Result

Thus, the given simple substitution ciphertext was successfully decrypted using **frequency analysis**.